

Amazon EC2 Auto Scaling Tutorial for Dynamic Scaling of Instances

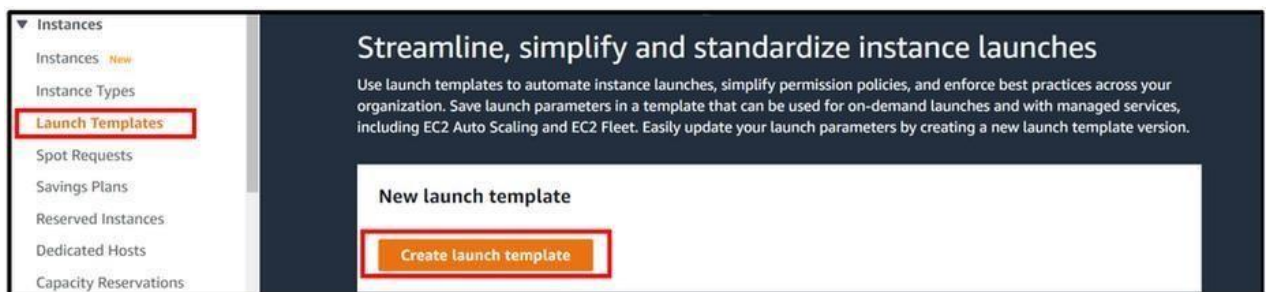
Dynamic Scaling of EC2 instance using Auto Scaling based on CPU Utilisation

Objectives:

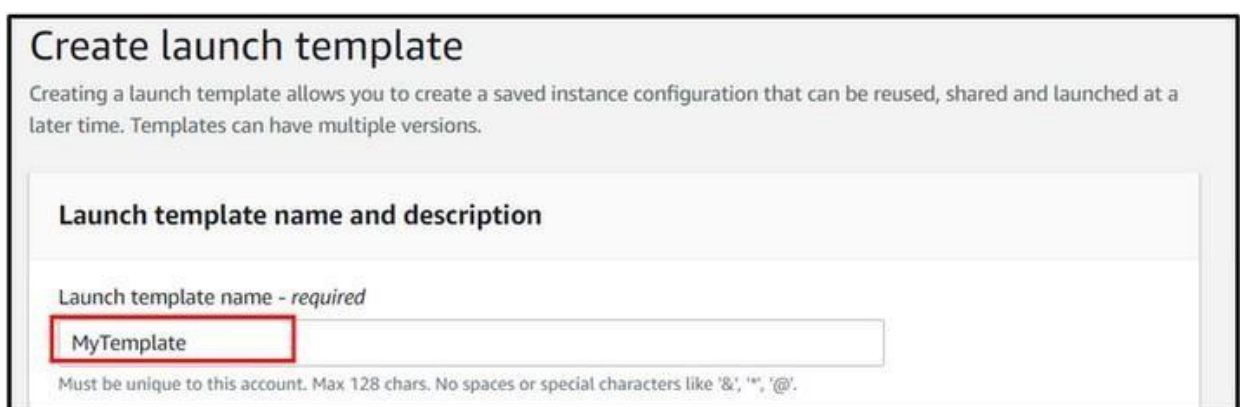
- Learn to create a launch template
- Learn to configure Auto Scaling group & create scaling policies for Amazon EC2 Auto Scaling.
- Learn to configure Amazon SNS for email notification.
- Learn to configure Amazon CloudWatch for monitoring CPU utilisation.

Step 1:

In **EC2** Service, on the left side panel go to **Instances -> Launch Template**. Click on **Create launch template**.



Name the template as **MyTemplate**.



Select **Amazon Machine Image (AMI)**: **Amazon Linux 2 AMI (HVM)**

Select **Instance Type** as **t2.micro**

Select your existing **Key Pair**

▼ Application and OS Images (Amazon Machine Image) [Info](#)

An AMI is a template that contains the software configuration (operating system, application server, and applications) required to launch your instance. Search or Browse for AMIs if you don't see what you are looking for below

 Search our full catalog including 1000s of application and OS images

Quick Start

Don't include
in launch
template

Amazon
Linux


Ubuntu


Windows


Red Hat


S



[Browse more AMIs](#)

Including AMIs from
AWS, Marketplace and
the Community

Amazon Machine Image (AMI)

Amazon Linux 2 AMI (HVM) - Kernel 5.10, SSD Volume Type
ami-08df646e18b182346 (64-bit (x86)) / ami-0e0aaf29e73155b91 (64-bit (Arm))
Virtualization: hvm ENA enabled: true Root device type: ebs

Free tier eligible ▼

Description

Amazon Linux 2 Kernel 5.10 AMI 2.0.20220606.1 x86_64 HVM gp2

Architecture

64-bit (x86) ▼

AMI ID

ami-08df646e18b182346

Verified provider

▼ Instance type [Info](#)

[Advanced](#)

Instance type

t2.micro
Family: t2 1 vCPU 1 GiB Memory
On-Demand Linux pricing: 0.0124 USD per Hour
On-Demand Windows pricing: 0.017 USD per Hour

Free tier eligible ▼

[Compare instance types](#)

▼ Key pair (login) [Info](#)

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name

MUM-KP ▼



[Create new key pair](#)

In **Network Settings** select the **Security Group** that is created for **Linux Server**. Keep the rest default values as it is.

▼ **Network settings** [Get guidance](#)

Subnet [Info](#)

Don't include in launch template ▼ [Create new subnet](#)

When you specify a subnet, a network interface is automatically added to your template.

Firewall (security groups) [Info](#)

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☒ Select existing security group ☐ Create security group

Security groups [Info](#)

Select security groups ▼ [Compare security group rules](#)

LinuxServer-SG X

► Advanced network configuration

Scroll down and click on **Create Launch Template**. Your Template is successfully created.

► Advanced network configuration

▼ **Storage (volumes)** [Info](#)

EBS Volumes [Hide details](#)

IOs, 1 GB of snapshots, and 100 GB of bandwidth to the internet.

Cancel **Create launch template**

Click on **View launch templates** at the bottom of new page and check whether it is created.

EC2 > Launch templates

Launch templates (1) [Info](#)

Filter by tags or properties or search by keyword

Launch template ID	Launch template name	Default version
☒ It-0d0ca46d26ffdeb40	MyTemplate	1

Step 2:

Now on the left side panel go to **Auto Scaling -> Auto Scaling Groups**. Click on **Create an Auto Scaling group**.

Name it as **MyASG**. Select the above created launch template **MyTemplate**.

Keep the rest default values as it is. Click on **Next** in bottom right corner.

Choose launch template or configuration [Info](#)

Specify a launch template that contains settings common to all EC2 instances that are launched by this Auto Scaling group. If you currently use launch configurations, you might consider migrating to launch templates.

Name

Auto Scaling group name
Enter a name to identify the group.

MyASG

Must be unique to this account in the current Region and no more than 255 characters.

Launch template [Info](#) [Switch to launch configuration](#)

Launch template
Choose a launch template that contains the instance-level settings, such as the Amazon Machine Image (AMI), instance type, key pair, and security groups.

MyTemplate

[Create a launch template](#)

Version
Default (1)

[Create a launch template version](#)

In **Configure Settings**, under **Network** click on the drop down arrow and select two subnets one at a time.

Network [Info](#)

For most applications, you can use multiple Availability Zones and let EC2 Auto Scaling balance across multiple Availability Zones. The default VPC and default subnets are suitable for getting started quickly.

VPC
Choose the VPC that defines the virtual network for your Auto Scaling group.

vpc-005bd2ca03ef65f49
172.31.0.0/16 Default

↻

[Create a VPC](#)

Availability Zones and subnets
Define which Availability Zones and subnets your Auto Scaling group can use in the chosen VPC.

Select Availability Zones and subnets

↻

ap-south-1a | subnet-059efbac9c29f0f79
172.31.32.0/20 Default

✕

ap-south-1b | subnet-08073cdc95e58f69d
172.31.0.0/20 Default

✕

[Create a subnet](#)

Keep the rest default values as it is. Click on **Next** in bottom right corner.

In **configure advanced options** keep the default values as it is. Click on **Next** in bottom right corner.

Under **Configure group size and scaling policies**, go to **Group size** and configure the values as: Desired Capacity: **2**

Minimum Capacity: **2**

Maximum Capacity: **5**

Configure group size and scaling policies [Info](#)

Set the desired, minimum, and maximum capacity of your Auto Scaling group. You can optionally dynamically scale the number of instances in the group.

Group size - optional [Info](#)

Specify the size of the Auto Scaling group by changing the desired capacity. You can also set minimum and maximum capacity limits. Your desired capacity must be within the limit range.

Desired capacity

Minimum capacity

Maximum capacity

In the **Scaling policies – optional**, select **Target tracking scaling policy** and configure the values as:

Scaling Policy Name: **MyASG Policy**

Metric Type: **Average CPU Utilization**

Target Value: **50**

Scaling policies - optional

Choose whether to use a scaling policy to dynamically resize your Auto Scaling group to meet changes in demand. [Info](#)

☒ Target tracking scaling policy
Choose a desired outcome and leave it to the scaling policy to add and remove capacity as needed to achieve that outcome.

☐ None

Scaling policy name

Metric type

Target value

Instances need
 seconds warm up before including in metric

☐ Disable scale in to create only a scale-out policy

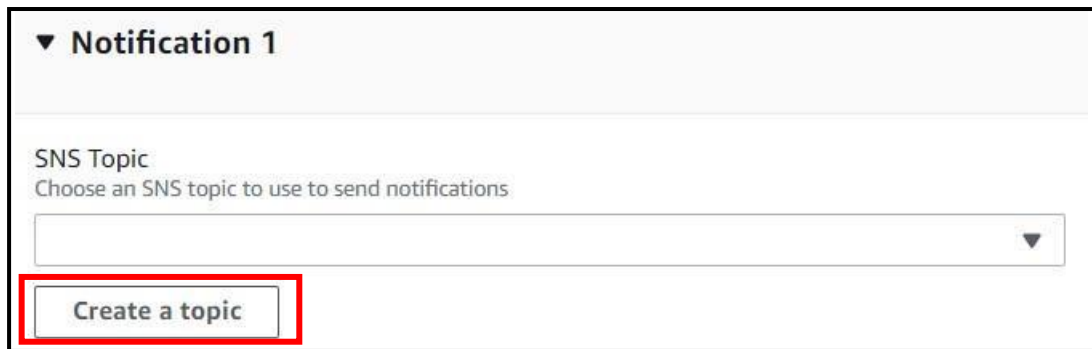
Click on **Next** in bottom right corner.

In **Add Notifications**, click on **Add notification**.

Add notifications [Info](#)

Send notifications to SNS topics whenever Amazon EC2 Auto Scaling launches or terminates the EC2 instances in your Auto Scaling group.

Click on **Create a Topic**



▼ Notification 1

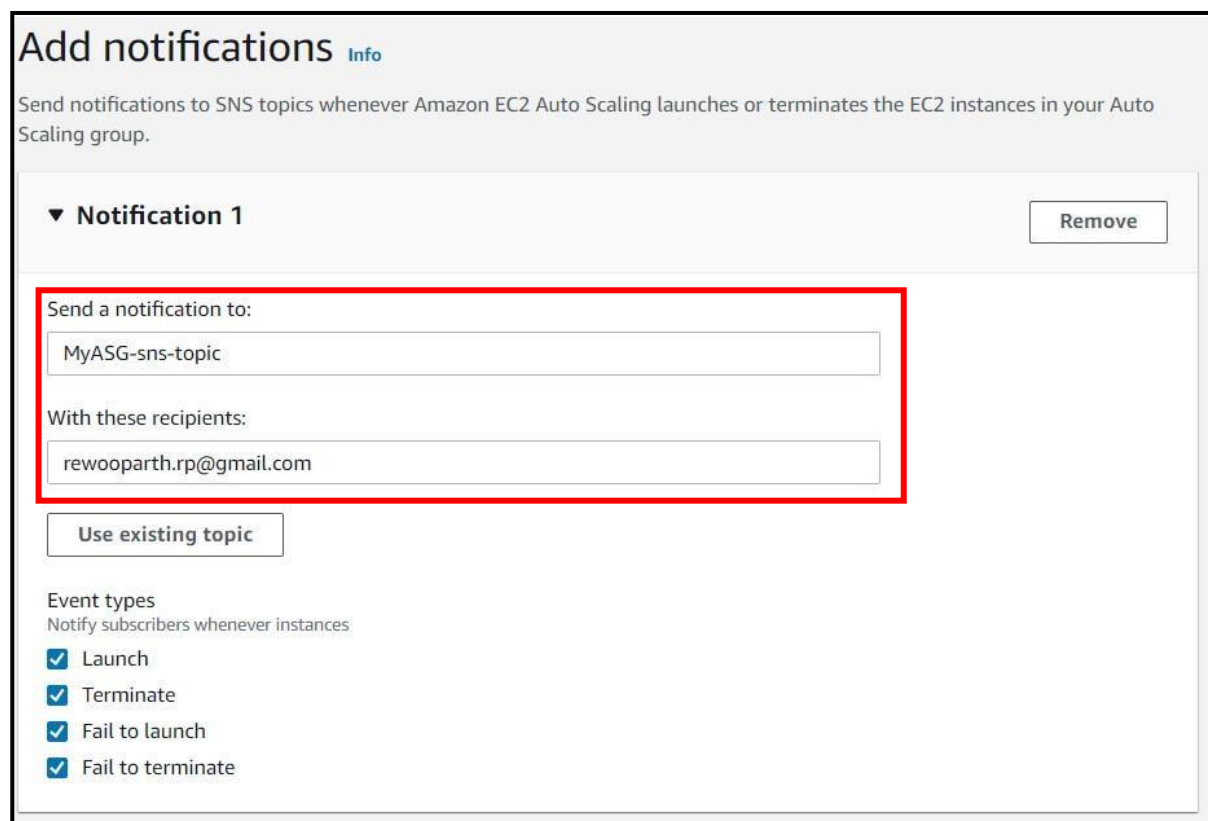
SNS Topic
Choose an SNS topic to use to send notifications

Create a topic

Configure the values as:

Send a notification to: MyASG-sns-topic

With these recipients: Enter your personal email address in this value



Add notifications [Info](#)

Send notifications to SNS topics whenever Amazon EC2 Auto Scaling launches or terminates the EC2 instances in your Auto Scaling group.

▼ Notification 1 Remove

Send a notification to:
MyASG-sns-topic

With these recipients:
rewooparth.rp@gmail.com

Use existing topic

Event types
Notify subscribers whenever instances

- ☒ Launch
- ☒ Terminate
- ☒ Fail to launch
- ☒ Fail to terminate

Click on **Next** In bottom right corner.

In **Add tags**, click on **Add tag** and configure the values as:

Key: Name

Value: EC2forASG

Add tags [Info](#)

Add tags to help you search, filter, and track your Auto Scaling group across AWS. You can also choose to automatically add these tags to instances when they are launched.

i You can optionally choose to add tags to instances (and their attached EBS volumes) by specifying tags in your launch template. We recommend caution, however, because the tag values for instances from your launch template will be overridden if there are any duplicate keys specified for the Auto Scaling group. ×

Tags (1)

Key	Value - optional	Tag new instances	
Name	EC2forASG	☒	Remove

49 remaining

Click on **Next** in bottom right corner.

Review your auto scaling group. After review, scroll down and click on **Create Auto Scaling group**.

Step 6: Add tags Edit

Tags (1)

Key	Value	Tag new instances
Name	EC2forASG	Yes

Your Auto Scaling group is successfully created. Check whether it is Created.

EC2 > Auto Scaling groups

Auto Scaling groups (1) Info

Search your Auto Scaling groups

Refresh Edit Delete Create an Auto Scaling group

☐	Name	Launch template/configuration	Instances	Status	Desired capacity	Min	Max	Availability Zones
☐	MyASG	MyTemplate Version Default	0	Updating capacity	2	2	5	ap-south-1b, ap-south...

The status would be initially **Updating Capacity**. This means that the given capacity is checked. The instances are being initiated or terminated accordingly.

With Auto Scaling Group you can monitor with use of these features. Thus, you can have better control created instances and the system.

Details	Activity	Automatic scaling	Instance management	Monitoring	Instance refresh
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Step 3:

On the left side panel go to **Instances -> View Instances**. Here you can see the working of Auto Scaling group to the specified capacity.

Instances (2) Info

Search

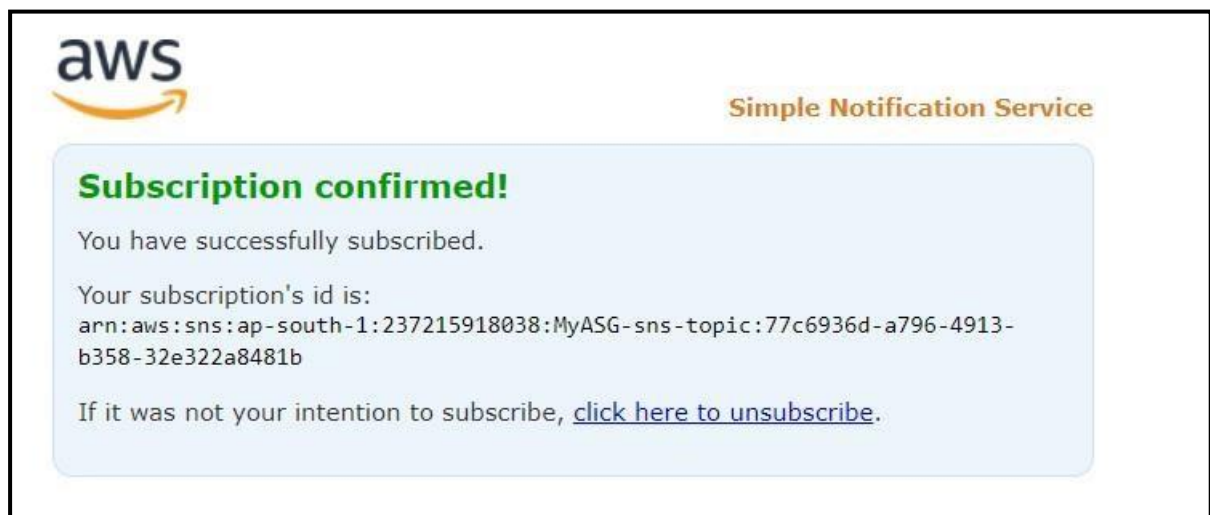
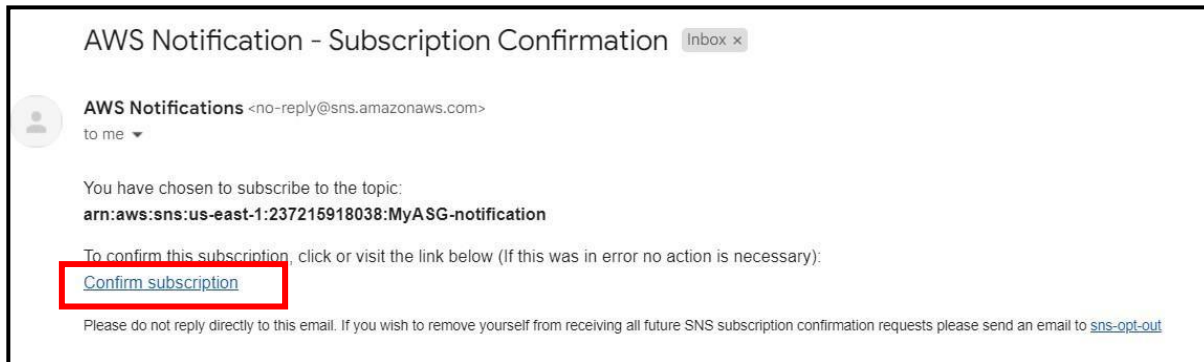
Refresh Connect Ins

☐	Name	Instance ID	Instance state	Instance type	Status check	Alarm status
☐	EC2forASG	i-07c28655813d7e8d9	Running	t2.micro	2/2 checks passed	No alarms
☐	EC2forASG	i-0dfa18c848aedc922	Running	t2.micro	2/2 checks passed	No alarms

Step 4:

Open your **Gmail** and check for the email subjected **AWS Notification - Subscription Confirmation**.

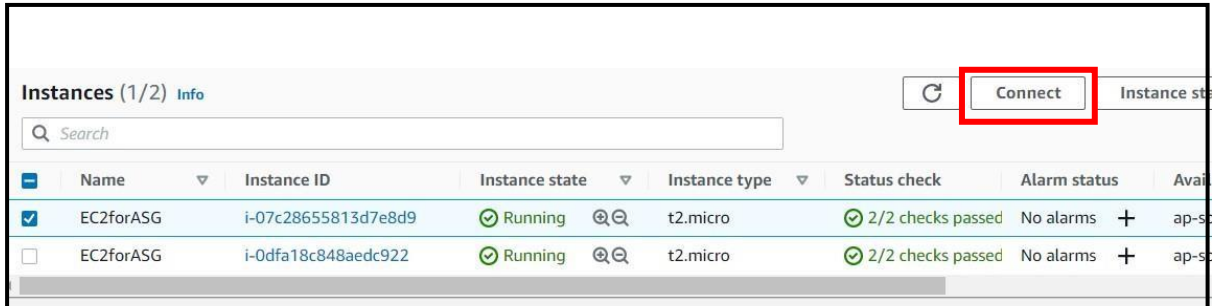
Open that email and click on **Confirm Subscription**.



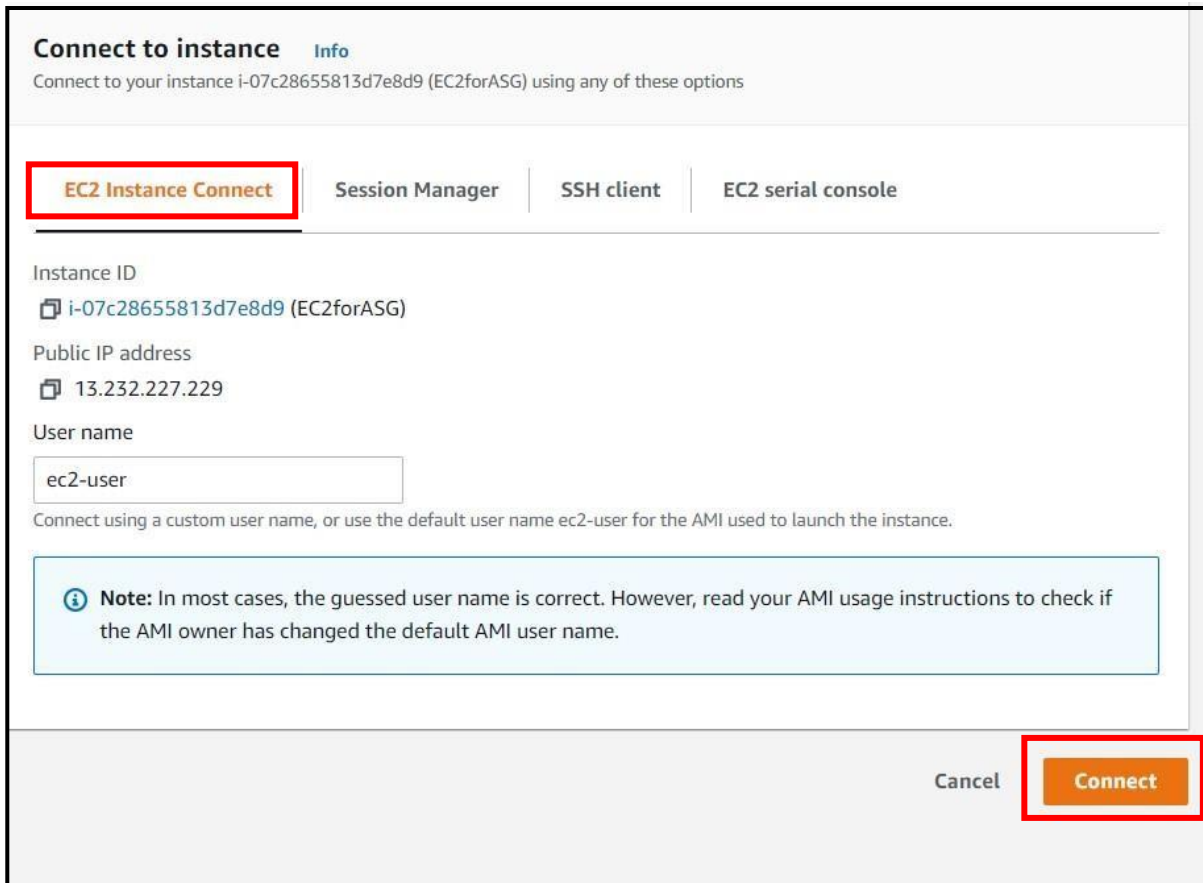
After confirming the subscription, close that tab and move back to the AWS console.

Step 5:

Connect to any one of the EC2 instance by **selecting the EC2 instance** and clicking the **Connect** button.



Select the **EC2 Instance Connect** and click on the **Connect** button in the right bottom corner.



Step 6:

In order to increase the **CPU Utilization** of the **EC2 instance**, we need to create a load on the server. This can be achieved by using **Stress**.

In order to use stress, we must be a **root user**. Run the following command to become a root user. `sudo -s`

```

  _ | _ | _ )
  _ | ( _ | /
  _ | \ _ | _ |
                                Amazon Linux 2 AMI

https://aws.amazon.com/amazon-linux-2/
12 package(s) needed for security, out of 22 available
Run "sudo yum update" to apply all updates.
[ec2-user@ip-172-31-38-169 ~]$ sudo -s
[root@ip-172-31-38-169 ec2-user]#
```

Now, we will install a utility called stress. This tool is designed to subject your system to a configurable measure of CPU, memory, I/O and disk stress.

Stress tools is not installed by default, you will need to install the package by using the following commands:

```
amazon-linux-extras install epel -y
```

```
yum install stress -y
```

Once installed, CPU load can be generated using Stress by running the following command:

```
stress --cpu 1 --timeout 300
```

The above command will take few minutes to complete.

```

[root@ip-172-31-38-169 ec2-user]# stress --cpu 1 --timeout 300
stress: info: [3995] dispatching hogs: 1 cpu, 0 io, 0 vm, 0 hdd
stress: info: [3995] successful run completed in 300s
[root@ip-172-31-38-169 ec2-user]#
```

After the above command is completed, go to Instances in the left side panel.

Refresh the page until you see some more EC2 instances created. *It may take some time.*

Instances (4) Info								
Search								
☐	Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	
☐	EC2forASG	i-02bd64711d6db4877	Running	t2.micro	Initializing	No alarms	ap-south-1a	+
☐	EC2forASG	i-07c28655813d7e8d9	Running	t2.micro	2/2 checks passed	No alarms	ap-south-1a	+
☐	EC2forASG	i-0dfa18c848aedc922	Running	t2.micro	2/2 checks passed	No alarms	ap-south-1b	+
☐	EC2forASG	i-046eb160bb90849f0	Running	t2.micro	2/2 checks passed	No alarms	ap-south-1b	+

Also, you can check your gmail in which you would have received emails subjected **Auto Scaling: launch for group "MyASG"**

It is the **SNS Notification trigger** which we have created while creating the Auto Scaling Group.

Auto Scaling: launch for group "MyASG"

Inbox x


AWS Notifications <no-reply@sns.amazonaws.com>
to me ▾

Service: AWS Auto Scaling
Time: 2022-08-01T20:38:39.568Z
RequestId: 3916096c-8565-0f52-0f5b-18ed78c5377a
Event: autoscaling:EC2_INSTANCE_LAUNCH
AccountId: 237215918038
AutoScalingGroupName: MyASG
AutoScalingGroupARN: arn:aws:autoscaling:ap-south-1:237215918038:autoScalingGroup:76932a21-a755-44c4-bf9d-6fc1dc708def
ActivityId: 3916096c-8565-0f52-0f5b-18ed78c5377a
Description: Launching a new EC2 instance: i-046eb160bb90849f0
Cause: At 2022-08-01T20:37:59Z a monitor alarm TargetTracking-MyASG-AlarmHigh-18d11171-1baa-47eb-a622-9407c4711285 in desired capacity from 2 to 4. At 2022-08-01T20:38:06Z an instance was started in response to a difference between desired and ac
StartTime: 2022-08-01T20:38:08.451Z
EndTime: 2022-08-01T20:38:39.568Z
StatusCode: InProgress

Move back to the Instance page. After some time, you will see that some of the EC2 instances are terminated by the Auto Scaling Group as the CPU Utilization of the application has lowered down.

Instances (4) Info							
Search							
☐	Name	Instance ID	Instance state	Instance type	Status check	Alarm status	
☐	EC2forASG	i-02bd64711d6db4877	Running	t2.micro	2/2 checks passed	No alarms	
☐	EC2forASG	i-07c28655813d7e8d9	Terminated	t2.micro	–	No alarms	
☐	EC2forASG	i-0dfa18c848aedic922	Shutting-down	t2.micro	–	No alarms	
☐	EC2forASG	i-046eb160bb90849f0	Running	t2.micro	2/2 checks passed	No alarms	

You will also receive email subjected **Auto Scaling: termination for group "MyASG"**

Auto Scaling: termination for group "MyASG" Inbox x



AWS Notifications <no-reply@sns.amazonaws.com>

to me ▾

Service: AWS Auto Scaling
Time: 2022-08-01T18:59:25.847Z
RequestId: 3f9a684e-9ca6-4530-a4e1-48a9ca730461
Event: autoscaling:EC2_INSTANCE_TERMINATE
AccountId: 237215918038
AutoScalingGroupName: MyASG
AutoScalingGroupARN: arn:aws:autoscaling:us-east-1:237215918038:autoScalingGroup:5af326
ActivityId: 3f9a684e-9ca6-4530-a4e1-48a9ca730461
Description: Terminating EC2 instance: i-0257b9f57f3cd2357
Cause: At 2022-08-01T18:58:10Z a monitor alarm TargetTracking-MyASG-AlarmLow-54a3f20a-3 the desired capacity from 4 to 3. At 2022-08-01T18:58:22Z an instance was taken out of service 3. At 2022-08-01T18:58:22Z instance i-0257b9f57f3cd2357 was selected for termination.

Step 7:

You can also watch the CPU Utilization graph on the CloudWatch Metrics.

Go to Left side panel and click **Auto Scaling Groups**.

Select **MyASG -> Monitoring -> EC2**.

The screenshot displays the AWS Management Console interface for the 'Auto Scaling groups' page. The breadcrumb navigation at the top shows 'EC2 > Auto Scaling groups'. The main heading is 'Auto Scaling groups (1/1) Info'. Below this is a search bar labeled 'Search your Auto Scaling groups'. A table lists the Auto Scaling groups, with the first row 'MyASG' highlighted and its checkbox selected. The table columns are 'Name', 'Launch template/configuration', 'Instances', 'Status', and 'De'. The 'MyASG' row shows 'MyTemplate | Version Default' for the launch template, '0' for instances, and 'Updating capacity' for status. Below the table is a navigation bar with tabs: 'Details', 'Activity', 'Automatic scaling', 'Instance management', 'Monitoring' (which is selected and highlighted with a red box), and 'Instance refresh'. Under the 'Monitoring' tab, the 'CloudWatch monitoring details' section shows two tabs: 'Auto Scaling' and 'EC2' (which is selected and highlighted with a red box). Below these tabs, there is a section for 'Auto Scaling group metrics collection' with an 'Enable' checkbox. A note states 'All times shown are in UTC.' and a link 'View all CloudWatch metrics' is provided. A '1h' time range selector is in the bottom right corner.

EC2 > Auto Scaling groups

Auto Scaling groups (1/1) Info

Search your Auto Scaling groups

☒	Name	Launch template/configuration	Instances	Status	De
☒	MyASG	MyTemplate Version Default	0	Updating capacity	2

Details | Activity | Automatic scaling | Instance management | **Monitoring** | Instance refresh

CloudWatch monitoring details

Auto Scaling | **EC2**

Auto Scaling group metrics collection:

☐ Enable

All times shown are in UTC.

[View all CloudWatch metrics](#)

1h

CloudWatch monitoring details

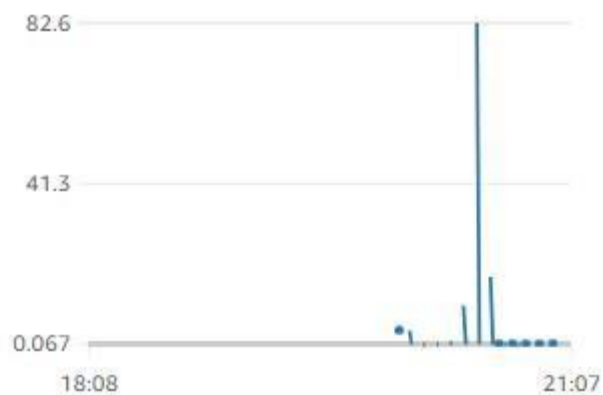
Auto Scaling

EC2

All times shown are in UTC.

[View all CloudWatch metrics](#) 

CPU Utilization (Percent)

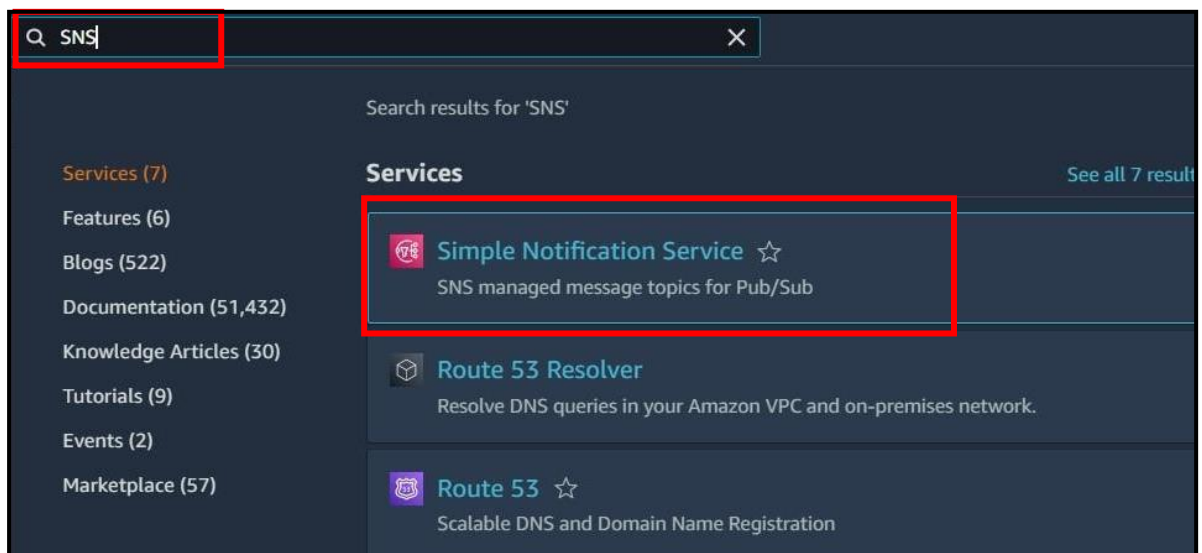


Note:

Delete the Template and the Auto Scaling Group if you do not need it anymore. This will also delete the instances created through the Auto Scaling Group.

Also delete the SNS Topic and Subscription created.

Search for SNS in AWS search bar and select Simple Notification Service.



Select Topics.



Select the Topic and click on Delete.



Similarly, do for Subscriptions.

Name	Version
Parth Rewoo	02-08-2022
Vaibhav Deshpande	03-08-2022